

Performance Analysis of InGaAs based Double Gate MOSFETs: Impact of Gate Dielectric Material and Thickness

ABSTRACT

As the scaling of conventional silicon-based MOSFETs approaches its physical limits, issues such as short-channel effects, increased leakage current, and degraded electrostatic control significantly affect device performance. These challenges restrict further improvement in speed and power efficiency, making it necessary to explore alternative materials and device architectures.

In this work, a comparative study of Double Gate (DG) MOSFETs based on silicon and III–V material (InGaAs) is carried out using Synopsys Sentaurus TCAD simulations. Initially, a silicon-based DG MOSFET is designed and analysed; however, the obtained performance is limited due to lower carrier mobility and weaker electrostatic control. To overcome these limitations, an InGaAs-based DG MOSFET is proposed and evaluated.

The transfer characteristics (I_{DS} – V_{GS}) of both devices are compared to demonstrate the superior performance of the InGaAs-based MOSFET in terms of higher ON-current and improved switching behaviour. Further, the impact of gate oxide thickness (T_{ox}) and dielectric material (k) on the performance of the InGaAs DG MOSFET is systematically analysed. Key performance parameters such as threshold voltage (V_T) and subthreshold swing (SS) are evaluated under varying T_{ox} and dielectric constants.

The results indicate that reducing oxide thickness enhances gate control, leading to improved device performance, while the use of high- k dielectric materials significantly improves electrostatic coupling and reduces subthreshold swing. Overall, the InGaAs-based DG MOSFET exhibits superior performance compared to the silicon-based device, making it a promising candidate for next-generation high-speed and low-power electronic applications.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid evolution of semiconductor technology has been primarily driven by the continuous scaling of Metal-Oxide-Semiconductor Field Effect Transistors (MOSFETs), which serve as the fundamental building blocks of modern electronic systems. Scaling has enabled increased device density, improved switching speed, and reduced power consumption, thereby supporting the development of compact and high-performance integrated circuits. However, as device dimensions approach the nanometer regime, conventional silicon-based MOSFETs encounter several critical challenges that limit further scaling and performance enhancement [4].

One of the major issues associated with scaled MOSFETs is the emergence of short-channel effects (SCEs), which degrade the electrostatic control of the gate over the channel [28]. This results in undesirable phenomena such as threshold voltage roll-off, drain-induced barrier lowering (DIBL), and increased leakage current. In addition, the reduction in gate oxide thickness leads to increased gate leakage, further contributing to power dissipation [1]. Another fundamental limitation is the subthreshold swing, which is thermally limited to 60 mV/dec at room temperature in conventional MOSFETs [10]. This restricts the ability to reduce supply voltage, thereby hindering the realization of ultra-low-power devices [27].

To overcome these limitations, advanced device architectures have been proposed, among which the Double Gate (DG) MOSFET is considered one of the most effective solutions. In a DG MOSFET, the channel is controlled by two gates placed on either side, providing superior electrostatic control compared to single-gate structures [6]. This configuration significantly suppresses short-channel effects, improves subthreshold characteristics, and enhances overall device performance [11]. The improved gate control also enables better scalability and reduced leakage current, making DG MOSFETs suitable for nanoscale applications [26].

Despite these structural improvements, the use of silicon as the channel material still imposes limitations due to its relatively low carrier mobility [1]. This has led to the exploration of alternative semiconductor materials, particularly III–V compounds, which exhibit superior transport properties. Among these materials, Indium Gallium Arsenide (InGaAs) has emerged as a promising candidate due to its high electron mobility, low effective mass, and enhanced carrier transport characteristics. These properties enable higher ON-state current and faster switching compared to silicon-based devices.

In this context, the present work focuses on the design and analysis of DG MOSFETs using both silicon and InGaAs channel materials. Initially, simulations are performed for a silicon-based DG MOSFET using Synopsys Sentaurus TCAD, and the results indicate limited performance improvement [10]. Subsequently, an InGaAs-based DG MOSFET is developed

to overcome these limitations [6]. A detailed comparative analysis is carried out using transfer characteristics (I_{DS} – V_{GS}) to highlight the superior performance of the InGaAs device. Furthermore, the influence of gate oxide thickness and dielectric material on device performance is investigated to understand their role in enhancing electrostatic control and switching behaviour.

1.2 Objective

The primary objective of this project is to design and analyze a Double Gate MOSFET using InGaAs as the channel material and to demonstrate its superior performance compared to a conventional silicon-based DG MOSFET. The study aims to highlight the advantages of combining advanced device architecture with high-mobility semiconductor materials for improved transistor performance.

A key objective of this work is to perform a comparative analysis of silicon and InGaAs DG MOSFETs by examining their transfer characteristics (I_{DS} – V_{GS}). This comparison focuses on evaluating the ON-state current, switching behaviour, and overall device efficiency [3]. Another important objective is to investigate the effect of gate oxide thickness on device performance, particularly its impact on threshold voltage and subthreshold swing [26]. Since oxide thickness directly influences gate control, its optimization is crucial for achieving better device characteristics.

In addition, the project aims to study the effect of different gate dielectric materials with varying dielectric constants [26]. The use of high- k dielectric materials is expected to improve gate capacitance and enhance electrostatic control over the channel, thereby reducing threshold voltage and improving subthreshold swing [30]. Through this analysis, the project seeks to establish the importance of material and structural optimization in modern transistor design.

Overall, the objective of this work is to demonstrate that the InGaAs-based DG MOSFET, combined with optimized oxide parameters, provides superior performance in terms of current conduction, switching characteristics, and electrostatic control compared to conventional silicon-based devices [13].

1.3 Report Organization

This report is systematically organized into five chapters to provide a clear and logical presentation of the work carried out in this project [3]. Each chapter focuses on a specific aspect of the study, starting from the basic concepts and progressing towards detailed analysis and conclusions.

Chapter 1 presents the introduction, which discusses the limitations of conventional silicon-based MOSFETs, the need for advanced device structures, and the motivation for using III–V materials such as InGaAs. It also outlines the objectives of the project and provides an overview of the report structure.

Chapter 2 provides a comprehensive literature survey, highlighting previous research work related to Double Gate MOSFETs, III–V semiconductor materials, and the impact of oxide

engineering on device performance [6]. This chapter establishes the background and research context for the present study [29].

Chapter 3 describes the simulation methodology used in this work. It includes details of the Synopsys Sentaurus TCAD tool, device structure design, simulation models, and parameter setup. This chapter explains how the silicon and InGaAs DG MOSFETs are modelled and analysed [21].

Chapter 4 presents the results and discussion, including the comparison of transfer characteristics between silicon and InGaAs devices [22]. It also includes a detailed analysis of the effect of gate oxide thickness and dielectric materials on threshold voltage and subthreshold swing [29]. The graphical results are used to justify the observed trends and performance improvements.

Chapter 5 concludes the report by summarizing the key findings of the study and emphasizing the advantages of InGaAs-based DG MOSFETs [25]. It also highlights the potential of these devices for future high-speed and low-power electronic applications.

1.4 Conclusion

In this chapter, the challenges associated with conventional silicon-based MOSFETs have been discussed, particularly in the context of nanoscale device scaling. The need for improved device architectures such as Double Gate MOSFETs and alternative semiconductor materials has been clearly established [18]. The advantages of using InGaAs as a channel material have been highlighted due to its superior transport properties and potential for enhanced device performance [4].

The objectives of the project have been defined, focusing on the design, simulation, and comparative analysis of silicon and InGaAs DG MOSFETs [15]. The organization of the report has also been outlined to provide a structured approach to presenting the work. This chapter lays the foundation for the subsequent chapters, which focus on literature review, simulation methodology, and detailed performance analysis of the proposed device.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The continuous scaling of MOSFET devices has been the driving force behind the advancement of modern electronics [19]. However, as device dimensions approach the nanometer regime, conventional design approaches face severe limitations [9]. To overcome these challenges, extensive research has been carried out in two major directions: structural engineering and material engineering [2]. Structural innovations such as multi-gate devices aim to improve electrostatic control, while material innovations such as III–V semiconductors aim to enhance carrier transport properties [24]. This chapter presents a detailed review of these approaches, highlighting their significance in improving MOSFET performance.

Moore's Law describes the historical trend in semiconductor technology where the number of transistors integrated on a single chip approximately doubles every 18 to 24 months, leading to continuous improvement in performance, reduction in cost per function, and increased device density [14]. This observation, originally made by Gordon Moore in 1965, has guided the semiconductor industry for several decades [21]. However, as device dimensions approach the nanometer scale, maintaining this trend has become increasingly challenging due to physical limitations such as short-channel effects, leakage currents, and power dissipation [10]. These challenges have motivated the exploration of advanced device structures like Double Gate MOSFETs and the use of alternative materials such as III–V semiconductors to sustain performance scaling beyond the limits of conventional silicon technology [12].

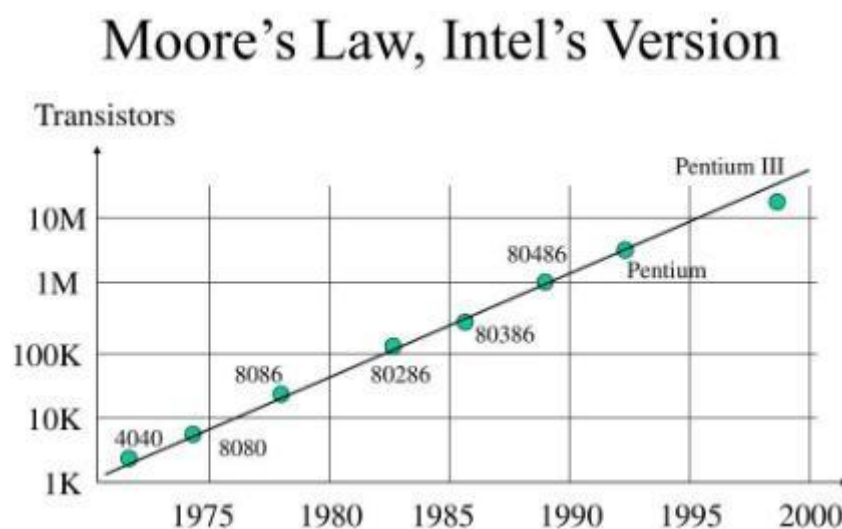


Fig.2.1: Moore's Law Trend in Intel Processors (1971-2000)

2.2 Conventional Single Gate MOSFET and Its Limitations

The conventional single gate MOSFET has been the backbone of semiconductor technology for several decades [28]. In this structure, a single gate controls the channel, modulating the current flow between source and drain. While this architecture has been highly successful at larger dimensions, it becomes increasingly ineffective as the channel length is reduced [2].

As scaling continues, short-channel effects become prominent, leading to degradation in device performance [27]. Researchers have reported that phenomena such as drain-induced barrier lowering (DIBL), threshold voltage roll-off, and increased leakage current become significant in nanoscale MOSFETs [9]. These effects arise due to the reduced ability of the gate to control the channel potential, allowing the drain voltage to influence the channel region.

Another critical limitation of the single gate MOSFET is the degradation of subthreshold characteristics. The subthreshold swing is fundamentally limited to 60 mV/dec at room temperature, which restricts further reduction in supply voltage [6]. This limitation directly impacts power consumption, making it difficult to design energy-efficient devices.

In addition, the reduction in oxide thickness required for scaling leads to increased gate leakage due to quantum tunneling. This further contributes to power dissipation and reliability issues [2]. These limitations have motivated researchers to explore alternative device structures that can provide better electrostatic control and improved performance.

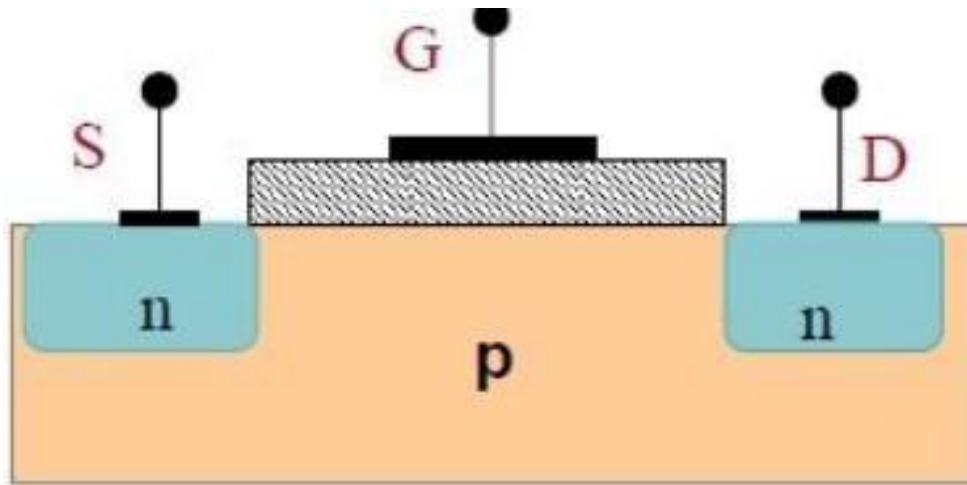


Fig.2.2: Cross - section of an n-channel MOSFET (NMOS)

2.3 Double Gate MOSFET as a Structural Solution

To overcome the limitations of single gate MOSFETs, multi-gate device architectures have been proposed, among which the Double Gate (DG) MOSFET is one of the most effective

solutions [20]. In a DG MOSFET, the channel is sandwiched between two gates, which simultaneously control the channel from both sides.

Research studies, including those by Taur and Ning, have demonstrated that DG MOSFETs provide superior electrostatic control compared to single gate devices [22]. The presence of two gates enhances the gate-to-channel coupling, which significantly reduces short-channel effects such as DIBL and threshold voltage variation [29]. This results in improved subthreshold characteristics and reduced leakage current.

The DG structure also enables better control over the channel potential, leading to a more uniform electric field distribution [13]. This improves carrier transport and enhances device performance [19]. Simulation-based studies using TCAD tools have confirmed that DG MOSFETs exhibit steeper subthreshold slopes and better scalability compared to conventional MOSFETs.

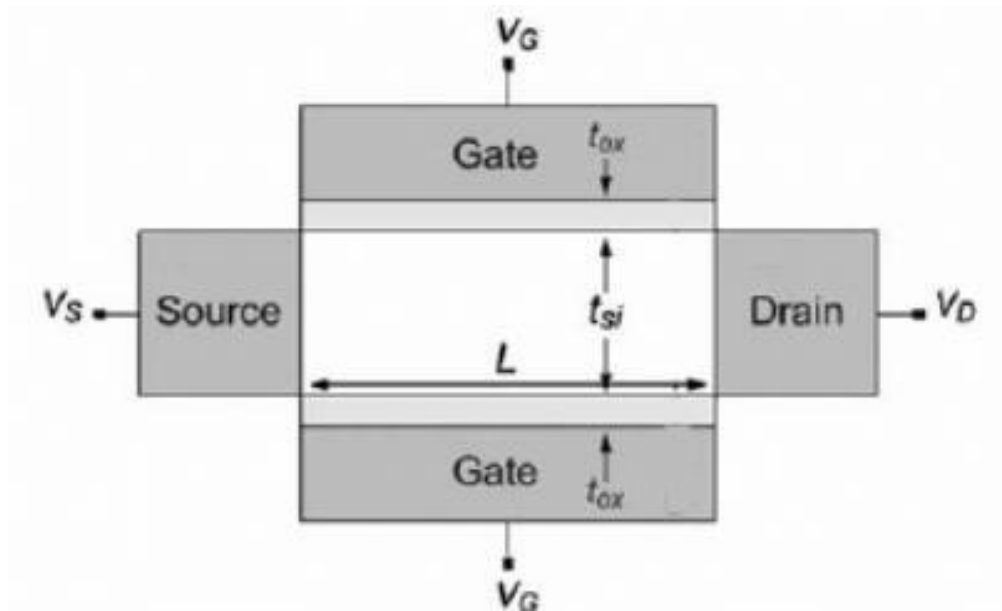


Fig2.3: Schematic of a Double-Gate(DG) MOSFET

Despite these advantages, the performance of DG MOSFETs is still influenced by the choice of channel material [3]. When silicon is used as the channel material, certain limitations persist, particularly in terms of carrier mobility and current drive capability. This has led to further research into alternative materials that can complement the advantages of DG structures [3].

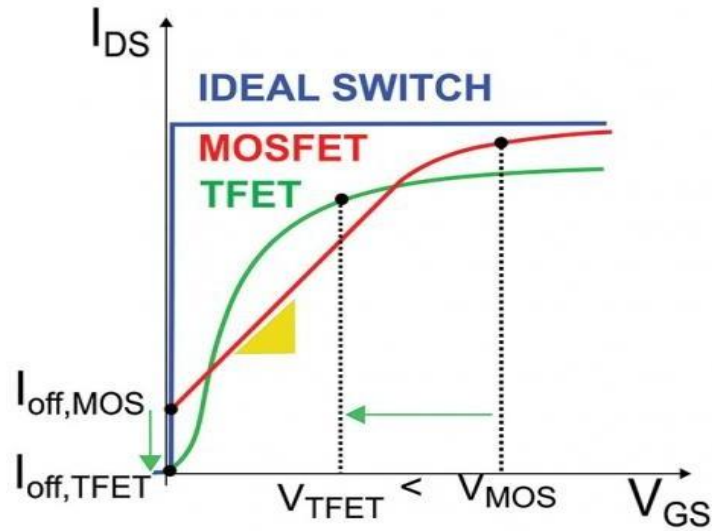


Fig.2.4: Typical transfer characteristics

2.4 Limitations of Silicon-Based MOSFETs

Silicon has been the dominant material for MOSFET fabrication due to its excellent material properties and well-established fabrication processes. However, as device dimensions shrink, the inherent limitations of silicon become more pronounced [21].

One of the primary limitations of silicon is its relatively low carrier mobility compared to III–V materials. This limits the ON-state current and reduces the switching speed of the device [17]. As a result, silicon-based MOSFETs struggle to meet the performance requirements of high-speed applications.

Additionally, silicon-based devices exhibit higher effective mass of charge carriers, which further restricts carrier transport [23]. This leads to increased channel resistance and reduced current conduction [7]. Researchers have also reported that silicon MOSFETs face challenges in maintaining strong inversion at very small dimensions, which affects device performance [25].

Another issue is the inability of silicon to provide sufficient performance improvement even with advanced structures such as DG MOSFETs. While structural engineering improves electrostatic control, the material limitations still restrict overall device performance [11]. This has motivated the exploration of alternative semiconductor materials with superior transport properties [18].

2.5 III–V Materials as a Material Engineering Solution

To overcome the limitations of silicon, researchers have explored III–V compound semiconductors such as InGaAs, GaAs, and InAs as alternative channel materials [10]. Among

these, InGaAs has gained significant attention due to its high electron mobility and excellent transport characteristics.

Studies by del Alamo et al. have shown that InGaAs-based MOSFETs exhibit significantly higher ON-state current and improved transconductance compared to silicon devices [8]. The high electron mobility in InGaAs allows for faster carrier transport, which results in improved switching speed and reduced delay.

The low effective mass of electrons in InGaAs also contributes to enhanced carrier injection and reduced scattering, leading to higher current conduction. These properties make InGaAs particularly suitable for high-frequency and high-speed applications.

However, the use of III–V materials also introduces challenges such as interface quality and fabrication complexity. Despite these challenges, research has shown that combining III–V materials with advanced device structures such as DG MOSFETs can significantly improve device performance. This combination represents a powerful approach to overcoming the limitations of conventional silicon-based devices.

2.6 Impact of Gate Oxide Thickness

Gate oxide thickness is a crucial parameter that directly affects the electrostatic control of the MOSFET [8]. Researchers have extensively studied the influence of oxide thickness on device performance, particularly in nanoscale devices.

A thinner oxide layer results in stronger gate control over the channel, which improves inversion and increases drain current [29]. This leads to better switching characteristics and reduced threshold voltage [7]. However, excessively thin oxide layers can result in increased gate leakage due to quantum tunneling, which negatively affects device reliability [12].

On the other hand, increasing the oxide thickness reduces the electric field at the channel, leading to weaker gate control. This results in higher threshold voltage and degraded subthreshold swing. These observations highlight the importance of optimizing oxide thickness to achieve a balance between performance and reliability.

2.7 High-k Dielectric Materials and Their Impact

To address the limitations associated with conventional SiO₂ gate oxide, high-k dielectric materials have been introduced in MOSFET design [14]. Materials such as HfO₂, Al₂O₃, and Si₃N₄ offer higher dielectric constants, which enable higher gate capacitance without requiring extremely thin oxide layers.

Research studies have shown that high-k dielectrics improve electrostatic control and reduce leakage current. The increased gate capacitance enhances the ability of the gate to control the channel, resulting in lower threshold voltage and improved subthreshold characteristics.

The use of high-k materials also allows for thicker physical oxide layers, which reduces gate leakage while maintaining high capacitance [15]. This makes high-k dielectrics an essential

component in modern MOSFET design [13]. When combined with DG structures and high-mobility materials, they further enhance device performance [11].

2.8 Combined Structural and Material Engineering Approach

Recent research trends focus on integrating both structural and material innovations to achieve optimal device performance. The combination of Double Gate MOSFET architecture with III–V materials such as InGaAs has been shown to provide significant improvements in device characteristics [1].

Studies indicate that DG MOSFETs with InGaAs channel material exhibit higher ON-current, improved subthreshold swing, and better electrostatic control compared to silicon-based devices [19]. Additionally, the incorporation of optimized oxide thickness and high-k dielectric materials further enhances device performance [6].

This combined approach represents a comprehensive solution to the challenges associated with nanoscale MOSFETs. It effectively addresses both structural and material limitations, making it a promising direction for future semiconductor device development [14].

2.9 Conclusion

This chapter reviewed the evolution of MOSFET technology from conventional single gate structures to advanced Double Gate architectures [15]. The limitations of silicon-based MOSFETs were discussed, highlighting the need for alternative materials. The advantages of III–V materials, particularly InGaAs, were presented as a solution to improve carrier transport and device performance [30].

The role of oxide engineering, including the optimization of oxide thickness and the use of high-k dielectric materials, was also discussed. Finally, the combined approach of structural and material engineering was highlighted as an effective strategy for enhancing MOSFET performance [24]. This forms the foundation for the present work, which focuses on the design and analysis of InGaAs-based DG MOSFETs.

CHAPTER 3

SIMULATION METHODOLOGY AND SOFTWARE DESCRIPTION

3.1 Introduction

The design and analysis of modern nanoscale semiconductor devices require advanced simulation tools capable of accurately modeling complex physical phenomena [23]. As device dimensions shrink to the nanometer regime, effects such as quantum confinement, carrier scattering, and non-uniform electric field distribution become increasingly significant. Experimental fabrication of such devices is both time-consuming and expensive, making simulation-based approaches essential for device development and optimization [11].

Technology Computer-Aided Design (TCAD) tools provide a platform for modeling semiconductor devices by solving fundamental physical equations that govern carrier transport [8]. These tools enable detailed analysis of electrical characteristics, allowing researchers to study the impact of material properties, device geometry, and operating conditions on performance.

In this work, Synopsys Sentaurus TCAD is employed to design and simulate Double Gate (DG) MOSFETs using silicon and InGaAs as channel materials. The simulation framework allows systematic comparison between the two devices and facilitates the analysis of key parameters such as threshold voltage and subthreshold swing [17]. Additionally, the influence of gate oxide thickness and dielectric materials is investigated using parameter variation techniques within the TCAD environment [30].

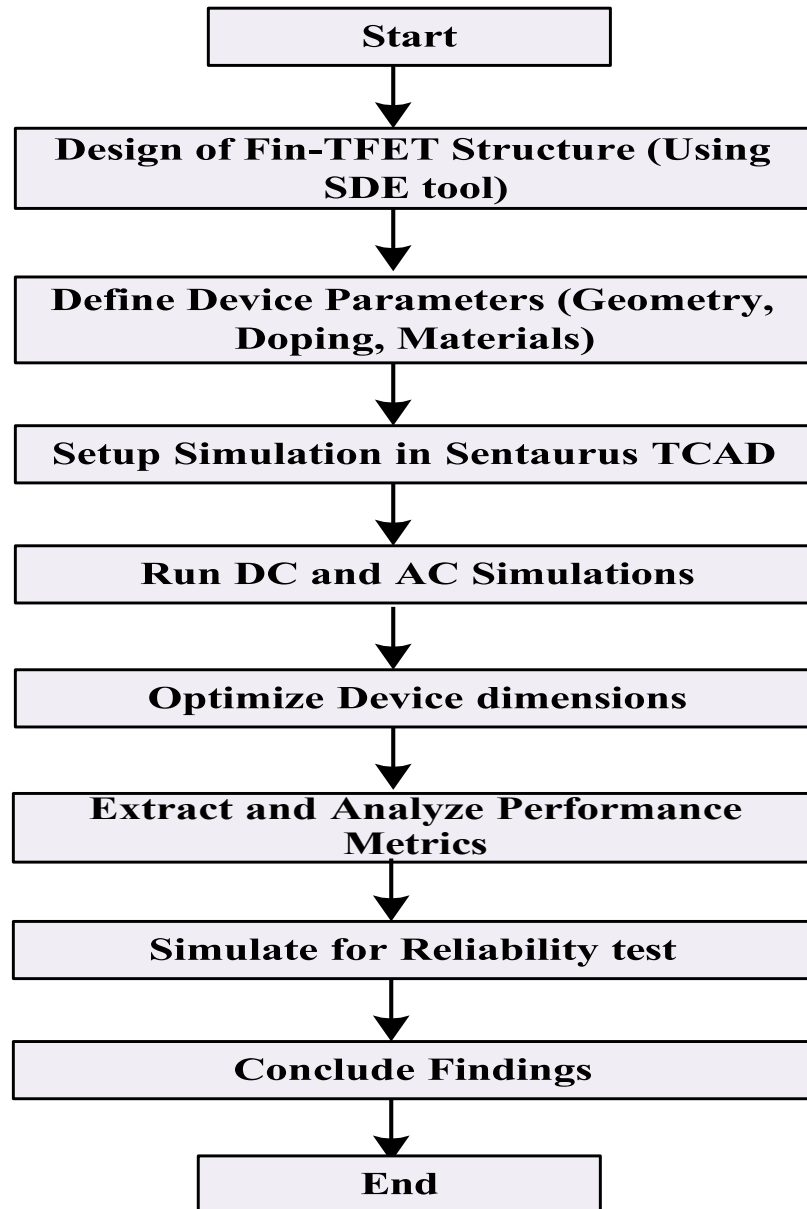


Fig.3.1: DG MOSFET Design flow

3.2 Overview of Synopsys Sentaurus TCAD

Synopsys Sentaurus TCAD is a comprehensive simulation platform widely used in both academic research and industrial applications for modeling semiconductor devices [25]. It provides a physics-based environment that enables the simulation of device behavior under various electrical, thermal, and material conditions.

The tool operates by solving a set of coupled differential equations, including Poisson's equation, continuity equations for electrons and holes, and transport models such as drift-diffusion [27]. For nanoscale devices, additional models such as quantum corrections and field-dependent mobility are incorporated to improve simulation accuracy.

One of the key advantages of Sentaurus TCAD is its modular structure, which allows users to separately define device geometry, physical models, and simulation conditions. This modularity makes it highly flexible and suitable for studying a wide range of semiconductor devices, including MOSFETs, TFETs, and heterojunction structures [4].

In this project, Sentaurus TCAD is used not only to simulate the electrical characteristics of DG MOSFETs but also to perform parameter sweeps and comparative analysis. The tool provides precise control over material properties such as bandgap, mobility, and dielectric constant, which is essential for modeling both silicon and InGaAs devices [23].

3.3 Sentaurus Workbench

Sentaurus Workbench (SWB) serves as the central control unit of the TCAD simulation environment. It provides an integrated platform where different simulation modules are connected in a sequential workflow [24]. The Workbench allows users to manage simulation projects, define dependencies between modules, and execute simulations efficiently.

In this work, Sentaurus Workbench is used to organize the simulation flow, starting from device structure creation to result visualization [11]. It enables the user to define input parameters such as channel material, oxide thickness, and dielectric constants, and automatically propagate these parameters across different simulation steps.

One of the important features of SWB is its ability to perform parameter sweeps [12]. This allows multiple simulations to be executed with varying values of parameters such as gate oxide thickness and dielectric constant [17]. Such capability is particularly useful in this study, where the impact of these parameters on threshold voltage and subthreshold swing is analysed systematically [28].

Furthermore, SWB provides a structured environment for comparing multiple device configurations, such as silicon and InGaAs DG MOSFETs. This ensures consistency in simulation conditions and enables accurate comparison of results.

3.4 Sentaurus Structure Editor (SDE)

Sentaurus Structure Editor (SDE) is used for defining the physical structure and geometry of the semiconductor device. It allows precise construction of device regions, assignment of materials, and specification of doping profiles.

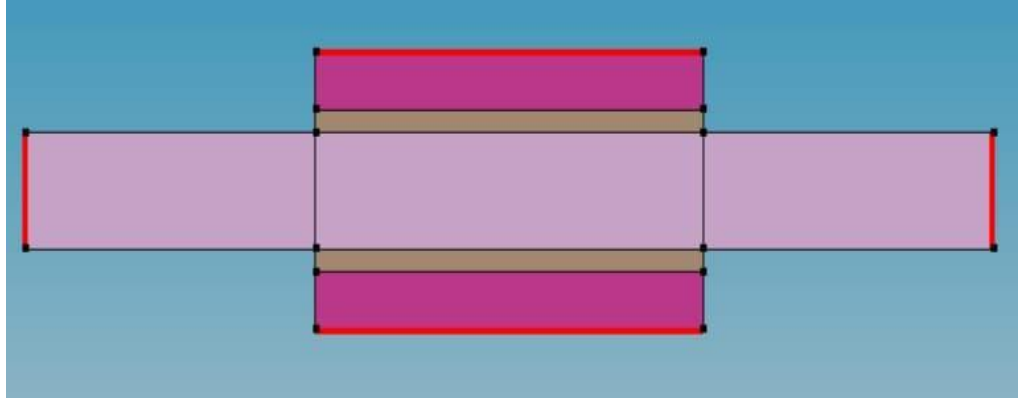


Fig.3.2: Sample design made with SDE tool.

In this work, DG MOSFET structures are designed using SDE for both silicon and InGaAs channel materials [17]. The device consists of source, drain, and channel regions, with the channel placed between two gate electrodes to form the double gate configuration [19]. The oxide layer is placed between the gate and channel, and its thickness is varied to study its effect on device performance.

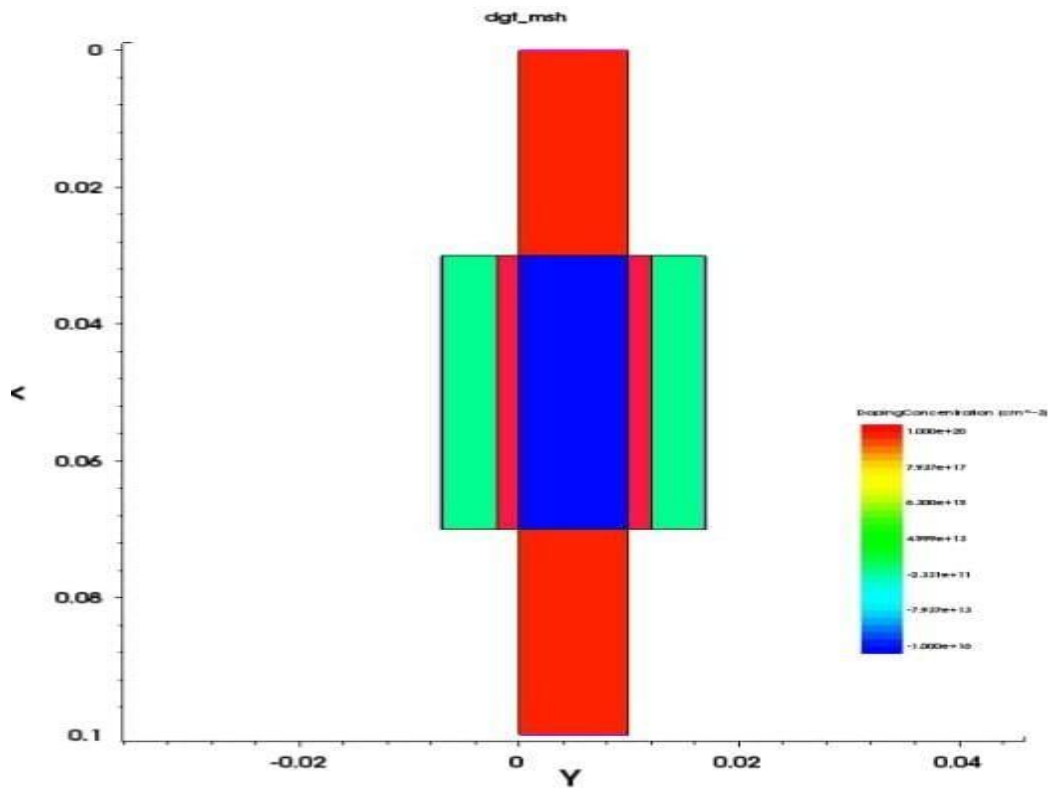


Fig.3.3: Model structure viewed using Sinspect tool.

The source and drain regions are heavily doped to ensure efficient carrier injection, while the channel is lightly doped to maintain proper control of the device. The use of appropriate doping profiles is essential for achieving realistic device behavior in simulations.

SDE also provides advanced meshing capabilities, which are critical for accurate simulation. A non-uniform mesh is employed, with finer mesh density near the channel and oxide interface, where electric field variations are significant. Proper mesh refinement ensures accurate calculation of carrier distribution and electric potential [22].

3.5 Sentaurus Device (SDEVICE)

Sentaurus Device (SDEVICE) is the core simulation module that performs electrical analysis of the device [20]. It solves the semiconductor transport equations to determine current-voltage characteristics and other electrical parameters.

In this study, SDEVICE is used to simulate the transfer characteristics (I_{DS} – V_{GS}) of both silicon and InGaAs DG MOSFETs. The simulation is carried out by applying appropriate bias conditions, where the gate voltage is swept while keeping the drain voltage constant.

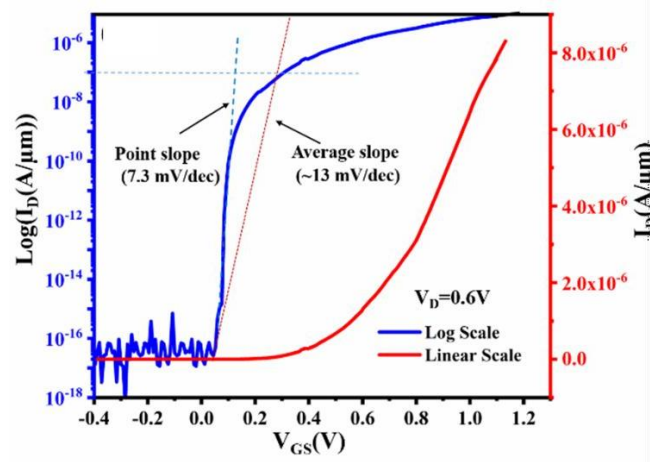


Fig.3.4: Model graph plotted using Sinspect tool.

To ensure accurate modeling, various physical models are incorporated, including mobility models, recombination-generation mechanisms, and quantum correction effects. These models account for phenomena such as carrier scattering, band structure effects, and carrier recombination, which are important for nanoscale devices [17].

The simulation results obtained from SDEVICE provide detailed information about device behavior, including current flow, electric field distribution, and carrier concentration [6]. These

results are used to extract key parameters such as threshold voltage, subthreshold swing, ON-current, and OFF-current [16].

Additionally, SDEVICE allows the inclusion of different dielectric materials and oxide thickness values, enabling the study of their impact on device performance [2].

3.6 Visualization Tools (Sentaurus Visual)

Sentaurus Visual is used to analyse and interpret the results obtained from device simulations. It provides graphical representations of electrical characteristics and spatial distributions of physical quantities.

In this work, Sentaurus Visual is used to plot IDS–VGS characteristics for both silicon and InGaAs DG MOSFETs. These plots are essential for comparing device performance and demonstrating the superiority of the InGaAs-based device [2].

The tool is also used to generate graphs showing the variation of threshold voltage and subthreshold swing with respect to oxide thickness and dielectric constant [27]. These graphs provide a clear understanding of how device performance is influenced by design parameters.

In addition to electrical characteristics, Sentaurus Visual can display spatial distributions such as electric field intensity and carrier concentration [25]. These visualizations help in understanding the internal behavior of the device and provide deeper insight into the physics governing device operation [20].

3.7 Simulation Methodology

The simulation methodology adopted in this work follows a systematic approach to ensure accurate and consistent analysis of DG MOSFET devices. Initially, the device structures are created using Sentaurus Structure Editor for both silicon and InGaAs channel materials [10]. Care is taken to maintain identical geometrical parameters for both devices to enable a fair comparison [17].

Once the structures are defined, the simulations are performed using Sentaurus Device. The transfer characteristics are obtained by sweeping the gate voltage over a specified range while maintaining a constant drain voltage [22]. This allows the extraction of key parameters such as threshold voltage and subthreshold swing [22].

Further analysis is carried out by varying the gate oxide thickness and dielectric material for the InGaAs-based DG MOSFET [30]. Multiple simulations are performed using different oxide thickness values to study their effect on device performance [17]. Similarly, different dielectric materials with varying dielectric constants are used to analyze their impact on electrostatic control.

3.8 Parameter Extraction

The extraction of key device parameters is essential for evaluating and comparing device performance. In this work, parameters such as threshold voltage, subthreshold swing, ON-current, and OFF-current are extracted from the simulated transfer characteristics [3].

The threshold voltage is determined from the I_{DS} – V_{GS} curve as the gate voltage at which the device transitions from the OFF state to the ON state [8]. The subthreshold swing is calculated from the slope of the I_{DS} – V_{GS} curve in the subthreshold region, which indicates how efficiently the device switches.

The ON-state current is measured at higher gate voltages, representing the maximum current flow through the device, while the OFF-state current is measured at low gate voltages, representing leakage current [6]. These parameters are used to compare the performance of silicon and InGaAs DG MOSFETs.

The extracted parameters also provide insight into the effect of oxide thickness and dielectric material on device performance [18]. This analysis helps in identifying optimal design conditions for improved MOSFET performance.

3.9 Conclusion

This chapter presented a detailed description of the simulation methodology and the tools used for analyzing DG MOSFET devices [8]. The functionalities of various Sentaurus TCAD modules, including Workbench, Structure Editor, Device, and Visual tools, were discussed in detail.

The methodology adopted for device design, simulation, and parameter extraction was explained, highlighting the systematic approach used in this work [20]. This chapter provides a strong foundation for the next chapter, which presents the simulation results and detailed performance analysis of silicon and InGaAs DG MOSFETs [4].

4.1 Introduction

This chapter presents a comprehensive analysis of the simulated Double Gate (DG) MOSFET devices using silicon and InGaAs as channel materials. The primary objective of this analysis is to evaluate the performance improvement achieved through material engineering and oxide optimization. The simulation results are obtained using Synopsys Sentaurus TCAD and are presented in the form of transfer characteristics and parameter variation graphs [11].

The comparison between silicon and InGaAs devices is carried out using IDS–VGS characteristics, which provide insight into current conduction and switching behaviour [28]. Further, the effect of gate oxide thickness and dielectric material on the performance of the InGaAs DG MOSFET is analysed in detail [24]. Parameters such as threshold voltage (V_T) and subthreshold swing (SS) are studied as functions of oxide thickness and dielectric constant [24].

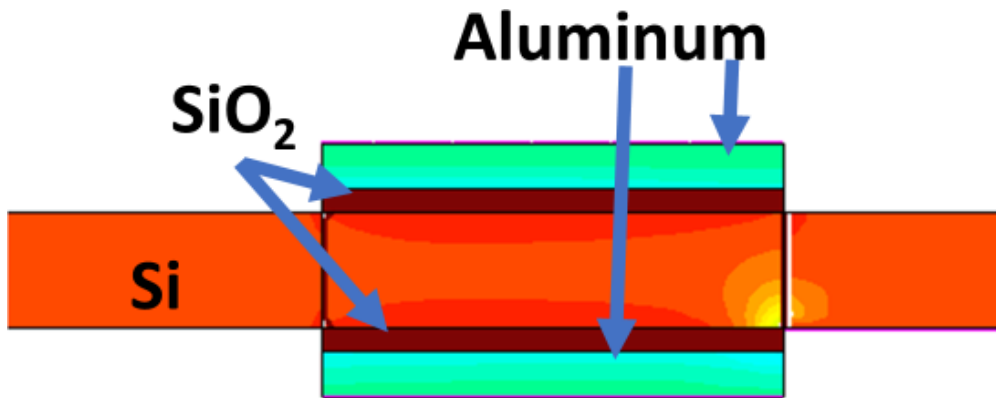


Fig.4.1: Structure of Double Gate MOSFET with silicon as channel material.

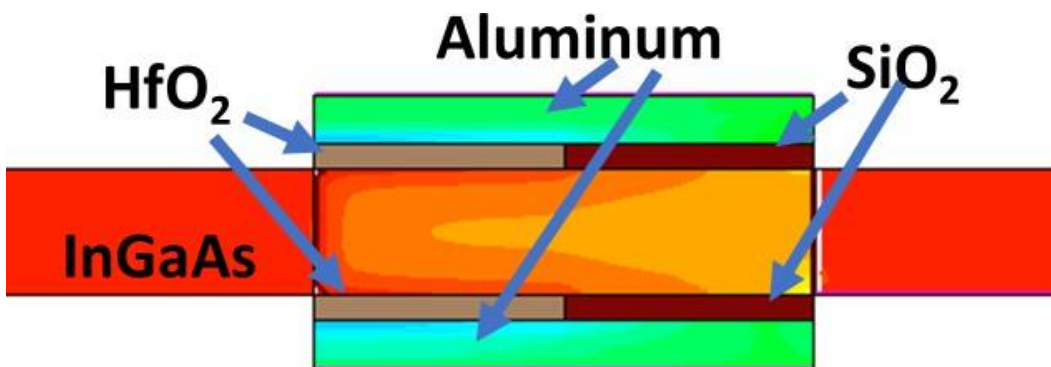


Fig.4.2: Structure of Double Gate MOSFET with InGaAs as channel material.

Table 4.1: Design specifications for DG MOSFET

Parameter	Value
Device Structure	Double Gate MOSFET
Channel Material	InGaAs / Silicon
Channel Length	40 nm
Source Length	30 nm
Drain Length	30 nm
Total Device Length	100 nm
Channel Thickness (Body Thickness)	10 nm
Gate Oxide Material	HfO ₂
Dielectric Constant (k)	~25
Gate Oxide Thickness (Tox)	1 nm
Gate Material	Metal Gate (e.g., TiN)
Source/Drain Doping	Heavily Doped (n ⁺)
Channel Doping	Lightly Doped / Intrinsic
Drain Voltage (Vd)	1 V
Gate Voltage (Vgs)	1 V
Simulation Tool	Synopsys Sentaurus TCAD

Each graph is analysed in detail to understand the observed trends, the underlying physical mechanisms, and their implications on device performance. This detailed analysis helps in identifying optimal design conditions for achieving high-performance MOSFET operation.

4.2 Transfer Characteristics of Si and InGaAs DG MOSFET

The I_{DS} – V_{GS} characteristics represent the fundamental behaviour of a MOSFET, illustrating how the drain current varies with the applied gate voltage [8]. From Fig.4.1, it is clearly observed that the InGaAs-based DG MOSFET exhibits a significantly higher drain current compared to the silicon-based device across the entire range of gate voltage. The ON-state current for the InGaAs device reaches the order of 10^{-4} A, whereas the silicon device shows

comparatively lower current levels. Both devices maintain a very low OFF-state current in the order of 10^{-14} A, indicating good leakage control; however, the transition from OFF to ON state is much sharper in the case of the InGaAs device [11].

The steeper slope of the I_{DS} – V_{GS} curve for the InGaAs MOSFET indicates improved switching characteristics and lower subthreshold swing. In contrast, the silicon device shows a relatively gradual increase in current, which reflects weaker gate control and slower switching behaviour [12]. Additionally, the threshold voltage for the InGaAs device appears lower, as the device begins to conduct at a smaller gate voltage compared to silicon [3]. This behaviour clearly demonstrates the advantage of using InGaAs as a channel material.

The superior performance of the InGaAs DG MOSFET can be explained by its intrinsic material properties [12]. InGaAs possesses a much higher electron mobility compared to silicon, which facilitates faster carrier transport in the channel [7]. The lower effective mass of charge carriers reduces scattering effects and enhances current conduction. Furthermore, the double gate structure provides strong electrostatic control, ensuring efficient channel formation [29]. Thus, it can be concluded that the combination of InGaAs material and DG structure results in significantly improved current drive capability and switching efficiency.

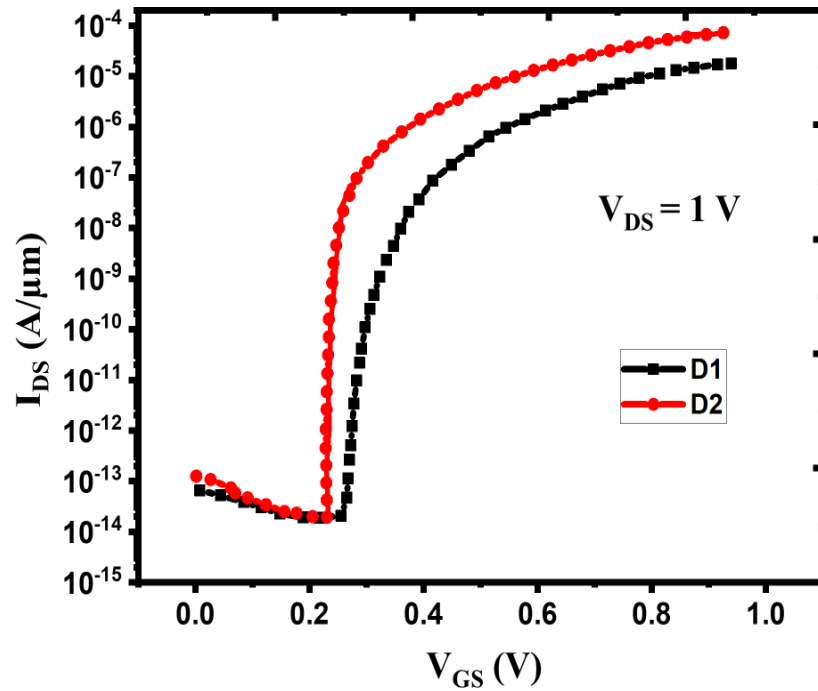


Fig.4.3: Transfer characteristics (I_{DS} – V_{GS}) of silicon-based and InGaAs-based DG MOSFET.

Electron density in two devices

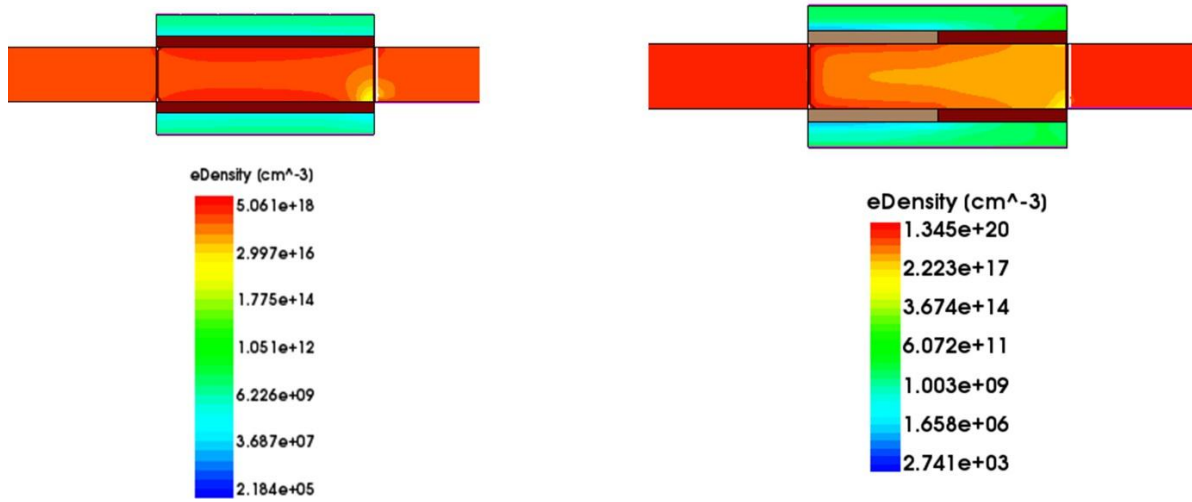


Fig.4.4: Contour plots for electron density for both the devices D1 and D2.

4.3 Effect of Oxide Thickness (T_{ox})

4.3.1 Threshold Voltage vs Oxide Thickness

The variation of threshold voltage with respect to oxide thickness is an important aspect in understanding the electrostatic behaviour of MOSFET devices [22]. From Fig. 4. 2, it is observed that the threshold voltage increases progressively as the oxide thickness increases from 1 nm to 4 nm [24]. The values range approximately from 0. 32 V for 1 nm oxide thickness to about 0. 47 V for 4 nm oxide thickness. This indicates that thinner oxide layers allow the device to turn ON at lower gate voltages, while thicker oxide layers require higher gate voltage to achieve the same level of conduction [6].

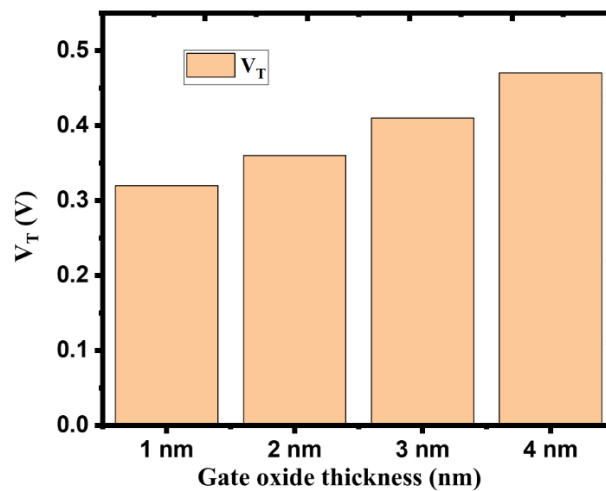


Fig.4.5(a): Impact of gate oxide thickness (T_{ox}) on the threshold voltage (V_T) of InGaAs DG MOSFET

$$T \propto \exp\left(-\frac{4\sqrt{2m^*E_G^3}}{3eh(E_G+\Delta\Phi)}\sqrt{\frac{\epsilon_{Si}}{\epsilon_{ox}}}\frac{t_{Si}}{t_{ox}}\right) \quad (2)$$

The observed trend can be explained based on the concept of gate capacitance and electric field coupling [27]. The gate capacitance is inversely proportional to the oxide thickness, and therefore, a thinner oxide layer results in higher capacitance. This increased capacitance enhances the electric field across the channel, enabling stronger control of the gate over the channel region [7]. As a result, the channel can be inverted at lower gate voltages, leading to a reduction in threshold voltage. On the other hand, as the oxide thickness increases, the gate capacitance decreases, weakening the electric field and reducing the gate's ability to control the channel [19]. This necessitates a higher gate voltage to achieve inversion.

In conclusion, the analysis clearly indicates that reducing oxide thickness is beneficial for lowering threshold voltage and improving device performance. However, it is also important to consider practical limitations such as gate leakage when choosing extremely thin oxide layers. Therefore, an optimal oxide thickness must be selected to balance performance and reliability.

4.3.2 Subthreshold Swing vs Oxide Thickness

The subthreshold swing (SS) is a critical parameter that determines how efficiently a MOSFET can transition from the OFF state to the ON state [8]. From Fig. 4. 3, it is observed that the subthreshold swing increases significantly with increasing oxide thickness, varying from approximately 70 mV/dec at 1 nm to about 102 mV/dec at 4 nm [30]. This indicates that the switching behaviour becomes less efficient as the oxide thickness increases.

$$SS_{Avg} = \frac{V_T - V_{OFF}}{\log(I_{ON}) - \log(I_{OFF})}$$

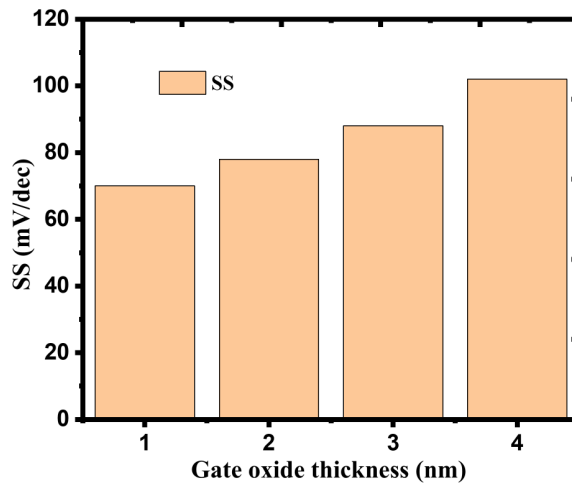


Fig.4.5(b): Variation of threshold voltage (V_T) with respect to gate oxide thickness (T_{ox}) for InGaAs DG MOSFET.

The increase in subthreshold swing can be attributed to the degradation of gate control over the channel. A thinner oxide layer allows a strong electric field to be established between the gate and the channel, resulting in rapid modulation of the channel potential [11]. This leads to a sharp increase in drain current with gate voltage, which is reflected as a lower subthreshold swing [8]. Conversely, a thicker oxide layer reduces the electric field strength, causing a slower response of the channel to changes in gate voltage. This results in a more gradual increase in drain current, leading to a higher subthreshold swing [15].

Thus, it can be concluded that minimizing oxide thickness is essential for achieving sharp switching characteristics and improved energy efficiency [22]. Lower subthreshold swing values are particularly important for low-power applications, as they enable faster switching at reduced operating voltages

4.3.3 I_{DS} – V_{GS} Characteristics for Different T_{ox}

This behaviour is primarily due to the enhanced gate control provided by thinner oxide layers [15]. A higher gate capacitance results in a stronger electric field, which facilitates efficient carrier injection into the channel [20]. This leads to higher carrier concentration and increased current conduction [22]. In contrast, thicker oxide layers reduce the effectiveness of the gate, resulting in weaker inversion and lower current levels.

In conclusion, the I_{DS} – V_{GS} characteristics confirm that reducing oxide thickness significantly improves current conduction and switching behaviour. This highlights the importance of oxide scaling in the design of high-performance MOSFETs [1].

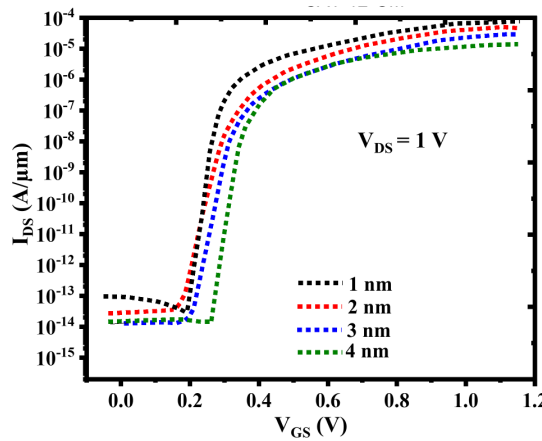


Fig.4.6: I_{DS} – V_{GS} characteristics of InGaAs DG MOSFET for different gate oxide thickness values.

This behaviour is primarily due to the enhanced gate control provided by thinner oxide layers. A higher gate capacitance results in a stronger electric field, which facilitates efficient carrier injection into the channel. This leads to higher carrier concentration and increased current conduction. In contrast, thicker oxide layers reduce the effectiveness of the gate, resulting in weaker inversion and lower current levels.

In conclusion, the I_{DS} – V_{GS} characteristics confirm that reducing oxide thickness significantly improves current conduction and switching behaviour. This highlights the importance of oxide scaling in the design of high-performance MOSFETs.

4.4 Effect of Dielectric Material (k)

4.4.1 Threshold Voltage vs Dielectric Constant

The threshold voltage variation with dielectric constant provides insight into the role of gate dielectric materials in MOSFET performance. From Fig. 4. 4, it is observed that the threshold voltage decreases as the dielectric constant increases, with values ranging from approximately 0.42 V for SiO_2 to about 0.30 V for HfO_2 . This indicates that high-k materials enable easier channel formation at lower gate voltages.

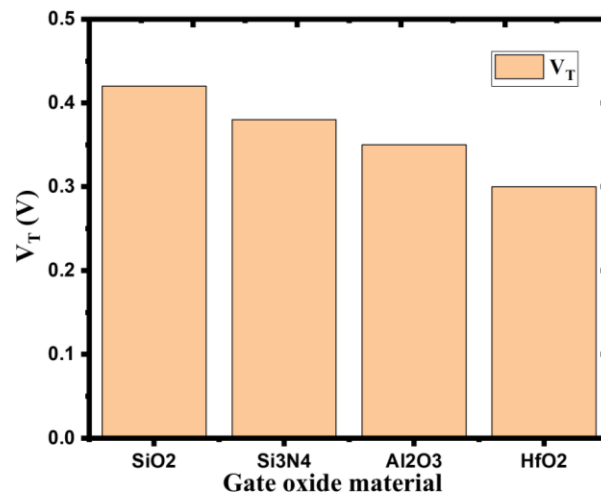


Fig.4.6(a): Variation of threshold voltage (V_T) with respect to dielectric constant (k) for different oxide materials in InGaAs DG MOSFET.

This trend can be explained by considering the relationship between dielectric constant and gate capacitance. A higher dielectric constant increases the capacitance of the gate oxide, which enhances the electric field across the channel. This improved electric field strengthens the gate's control over the channel, allowing inversion to occur at lower gate voltages. In contrast, low-k materials such as SiO_2 provide weaker gate control, requiring higher gate voltage to achieve the same effect.

Thus, it can be concluded that high-k dielectric materials are highly beneficial for reducing threshold voltage and improving device efficiency [17]. Their use is particularly important in advanced nanoscale devices.

4.4.2 Subthreshold Swing vs Dielectric Constant

From Fig.4.5, it is observed that the subthreshold swing decreases as the dielectric constant increases, with values ranging from approximately 95 mV/dec for SiO_2 to about 68 mV/dec for

HfO₂ [5]. This indicates a significant improvement in switching behaviour with the use of high-k materials.

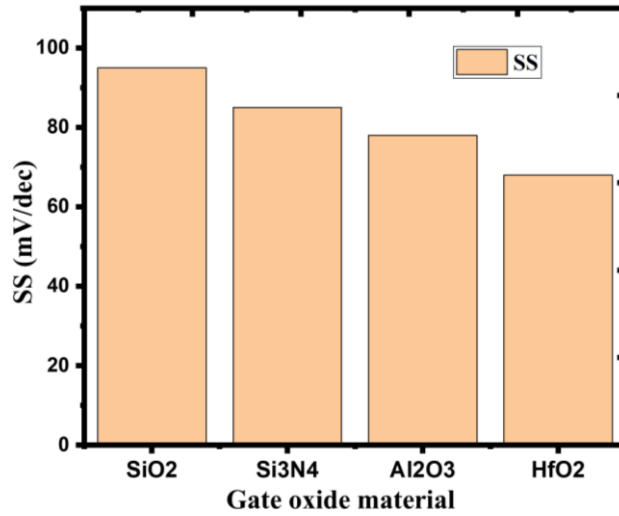


Fig.4.6(b): Variation of subthreshold swing (SS) with respect to dielectric constant (k) for different oxide materials in InGaAs DG MOSFET.

The reduction in subthreshold swing is due to the enhanced gate control provided by higher dielectric constant materials. Increased capacitance enables stronger modulation of the channel potential, allowing the drain current to increase rapidly with gate voltage. This results in a steeper I_{DS} – V_{GS} curve in the subthreshold region.

In conclusion, high-k dielectric materials play a crucial role in improving switching efficiency and reducing power consumption, making them essential for modern MOSFET design [10].

4.4.3 I_{DS} – V_{GS} Characteristics for Different Dielectric Materials

The I_{DS} – V_{GS} characteristics for different dielectric materials show that devices with higher dielectric constants exhibit higher ON-state current and improved switching behaviour. Materials such as HfO₂ result in ON-current in the order of 10^{-4} A, while SiO₂ shows comparatively lower current levels.

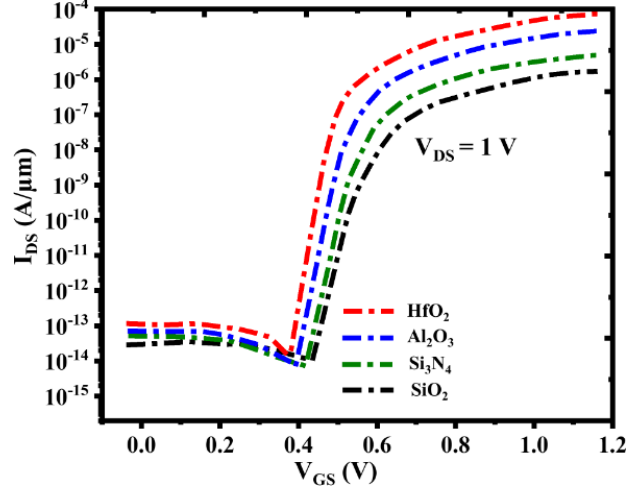


Fig.4.7: I_{DS} – V_{GS} characteristics of InGaAs DG MOSFET for different dielectric materials (varying k values).

This behaviour is due to the increased gate capacitance associated with high- k materials, which enhances the electric field at the channel and improves carrier injection. As a result, the device achieves stronger inversion and higher current conduction.

Thus, it can be concluded that the use of high- k dielectric materials significantly enhances the overall performance of DG MOSFETs.

4.5 Comparative Analysis of Si and InGaAs DG MOSFET

The comparative analysis of silicon-based and InGaAs-based Double Gate MOSFETs is essential to clearly understand the advantages offered by material engineering in nanoscale devices [28]. While both devices are designed with identical geometrical parameters and simulated under similar conditions, their performance differs significantly due to intrinsic material properties [15].

From the simulation results, it is evident that the InGaAs DG MOSFET exhibits superior electrical characteristics compared to the silicon-based device [30]. The most prominent improvement is observed in the ON-state current (I_{ON}), where the InGaAs device achieves current levels in the order of 10^{-4} A, whereas the silicon device shows comparatively lower current. This enhancement is primarily due to the higher electron mobility and lower effective mass of carriers in InGaAs, which facilitate faster carrier transport and improved current conduction [5].

In terms of OFF-state current (I_{OFF}), both devices maintain very low leakage current in the order of 10^{-14} A, indicating effective suppression of leakage in the DG structure. However, the InGaAs device achieves this low leakage while simultaneously providing higher ON-current, resulting in a better overall switching performance [30].

The subthreshold swing (SS) is another critical parameter that highlights the switching efficiency of the device. The InGaAs MOSFET exhibits a lower SS compared to the silicon

device, indicating a sharper transition from OFF to ON state. This improvement is attributed to stronger electrostatic control and enhanced carrier response in the InGaAs channel [7].

Similarly, the threshold voltage (V_T) of the InGaAs device is observed to be lower than that of the silicon device. This indicates that the InGaAs MOSFET requires a lower gate voltage to turn ON, which is advantageous for low-power operation. The reduced threshold voltage is a result of improved gate control and higher carrier injection efficiency [7].

Overall, the comparative analysis clearly demonstrates that the InGaAs-based DG MOSFET outperforms the silicon-based device in terms of current conduction, switching efficiency, and operating voltage. This highlights the effectiveness of combining advanced device architecture with high-mobility materials for next-generation semiconductor applications.

Table 4.2: Comparison of Si and InGaAs DG MOSFET

Parameter	Si DG MOSFET	InGaAs DG MOSFET
ON Current (I_{ON})	2×10^{-5} A	3.1×10^{-4} A
OFF Current (I_{OFF})	5.4×10^{-14} A	2×10^{-14} A
Threshold Voltage (V_T)	0.42 V	0.30 V
Subthreshold Swing (SS)	90–100 mV/dec	65–75 mV/dec
Switching Behaviour	Moderate	Sharp and efficient
Carrier Mobility	Low	High
Overall Performance	Limited	Superior

4.6 Conclusion

In this chapter, a detailed analysis of the performance of Double Gate MOSFETs based on silicon and InGaAs has been presented [13]. The transfer characteristics clearly demonstrated that the InGaAs-based device exhibits significantly higher ON-state current while maintaining low OFF-state current, indicating improved current conduction without compromising leakage performance [7].

The effect of gate oxide thickness and dielectric material on device performance was also analysed [14]. It was observed that reducing oxide thickness enhances gate control, leading to lower threshold voltage and improved subthreshold swing [5]. Similarly, the use of high-k dielectric materials improves electrostatic coupling between the gate and the channel, resulting in better switching characteristics and reduced operating voltage [23].

The I_{DS} – V_{GS} characteristics under varying oxide thickness and dielectric materials further confirmed that optimized oxide engineering plays a crucial role in enhancing device performance. Thinner oxide layers and high-k materials result in higher ON-current and sharper switching behaviour.

Finally, the comparative analysis between silicon and InGaAs DG MOSFETs clearly established the superiority of the InGaAs-based device. The improved carrier mobility and enhanced electrostatic control make it a promising candidate for future high-speed and low-power electronic application.

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

In this project, a comprehensive study of Double Gate (DG) MOSFETs based on silicon and InGaAs channel materials has been carried out using Synopsys Sentaurus TCAD simulations. The primary objective was to investigate the impact of material engineering and oxide optimization on device performance and to demonstrate the superiority of InGaAs-based DG MOSFET over conventional silicon-based devices [24].

Initially, a silicon-based DG MOSFET was designed and simulated to analyse its electrical characteristics. Although the double gate structure improved electrostatic control compared to conventional MOSFETs, the overall performance was limited due to the intrinsic material properties of silicon, particularly its lower carrier mobility [9]. The obtained results indicated moderate ON-state current and relatively slower switching behaviour [28].

To overcome these limitations, an InGaAs-based DG MOSFET was designed and analysed under similar conditions [2]. The simulation results clearly demonstrated that the InGaAs device exhibits significantly improved performance. The transfer characteristics showed a higher ON-state current in the order of 10^{-4} A, while maintaining a very low OFF-state current in the order of 10^{-14} A. This indicates that the device achieves high current conduction without compromising leakage performance [5].

Further analysis was carried out to study the effect of gate oxide thickness and dielectric material on the performance of the InGaAs DG MOSFET. It was observed that reducing oxide thickness enhances gate control, resulting in lower threshold voltage and improved subthreshold swing. Similarly, the use of high-k dielectric materials significantly improves electrostatic coupling between the gate and the channel, leading to sharper switching characteristics and reduced operating voltage [13].

The $I_{DS}-V_{GS}$ characteristics for varying oxide thickness and dielectric materials further confirmed that device performance is highly dependent on oxide engineering [12]. Thinner oxide layers and high-k materials resulted in improved current conduction and switching efficiency [19]. These findings highlight the importance of optimizing both structural and material parameters in MOSFET design.

Overall, the comparative analysis clearly established that the InGaAs-based DG MOSFET outperforms the silicon-based device in terms of ON-current, threshold voltage, and subthreshold swing. The superior carrier mobility and enhanced electrostatic control make InGaAs a highly suitable material for next-generation high-speed and low-power electronic devices.

5.2 Future Scope

Although the present work demonstrates significant improvement in DG MOSFET performance using InGaAs material, there are several opportunities for further research and enhancement [22].

One possible extension of this work is the exploration of other III–V materials such as InAs and GaSb, which offer even higher carrier mobility and can further improve device performance [18]. These materials can be used in combination to form heterojunction devices, which may provide additional advantages in terms of carrier transport and switching efficiency [7].

Another important direction is the investigation of advanced device architectures such as Gate-All-Around (GAA) MOSFETs and FinFETs [6]. These structures provide even better electrostatic control compared to DG MOSFETs and are widely considered for future technology nodes.

Further work can also be carried out on optimizing device parameters such as channel length, doping concentration, and gate work function to achieve improved performance. The impact of scaling these parameters on device reliability and variability can also be studied [30].

In addition, the use of advanced dielectric materials and interface engineering techniques can be explored to further reduce leakage current and improve device stability. The study of interface traps and their effect on device performance can also be an important area of research [9].

Finally, the integration of InGaAs-based MOSFETs into circuit-level applications such as logic circuits and analog systems can be investigated to evaluate their practical feasibility [29]. Experimental validation and fabrication-based studies can also be pursued to verify the simulation results and address real-world challenges [5].

5.3 Final Remark

The results obtained in this work clearly indicate that the combination of advanced device structures and high-mobility materials is essential for overcoming the limitations of conventional MOSFETs [14]. The InGaAs-based DG MOSFET demonstrates strong potential for future semiconductor technologies, particularly in applications requiring high speed and low power consumption [

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